THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Full Syllabus (Chapters 9–14) | Class 9 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(a)** both the proposition and its converse are true	the square of an odd number is odd
2	**(d)** 7	it satisfies the 'if' part but not the 'then' part
3	**(b)** 61	$\frac{20\times 55+30\times 65}{20+30}=61$
4	**(d)** 12	$\frac{6\cdot 4+10y}{6+y}=8\Rightarrow y=12$
5	**(a)** adding two single-digit numbers	algorithms are built from basic steps we can already perform
6	**(b)** 6	divisors(18) = [1, 2, 3, 6, 9, 18]
7	**(a)** no three collinear, all in one plane, and non-self-intersecting sides	Definition 1 of the chapter
8	**(c)** 9 cm	PQ = BC/2
9	**(c)** 2	rise ÷ run
10	**(d)** (1, 2)	substitute each point in the equation
11	**(c)** 248 cm^2	$2(lw+wh+lh)=2(60+24+40)=248$
12	**(d)** 550 cm^2	$\pi rl=550$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	Combined average $=\frac{an+bm}{n+m}$ (2).	2
14	All $j=1,2,\ldots,n$ — that is n numbers (1). We test j in increasing order and append at the end, so the list stays in increasing order (1).	1+1
15	If three vertices are collinear, two adjacent sides lie along one line (1), giving a triangle or a figure with overlapping sides instead of a quadrilateral (1).	1+1
16	Substituting $x=0,\ y=0$ gives $0-0=0$ (1). So $(0,0)$ satisfies the equation and lies on the line; this happens whenever $c=0$ (1).	1+1
17	A cone has one-third the volume of the cylinder with the same base and height (1). Fill the cone with salt and empty it into the cylinder; it takes exactly three fills (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Write $m=ad$, $n=bd$ for natural numbers a,b (1). $m-n=ad-bd=(a-b)d$ (1). Hence d divides $m-n$ (1).	1+1+1
19	$AD\parallel BC\Rightarrow\angle DAC=\angle ACB$; $AB\parallel DC\Rightarrow\angle BAC=\angle ACD$ (alternate angles) (2). As $\angle BAC=\angle DAC$, we get $\angle ACB=\angle ACD$, so AC bisects $\angle C$ (1).	2+1
20	Slope $=\frac{7-3}{3-1}=2$ (1). Using $(1,3)$: $3=2(1)+d$, so $d=1$ (1). Equation: $y=2x+1$; check $(3,7)$: $2(3)+1=7$ (1).	1+1+1
21	$V=\frac{1}{3}\times 81\times 10=270$ cm^3 (2). Prism volume $=810$ cm^3, so the pyramid is one-third of it (1).	2+1
22	Sum of ratio parts $=12\Rightarrow$ one part $=30^\circ$ (1). Angles: $60^\circ,\ 90^\circ,\ 90^\circ,\ 120^\circ$ (2).	1+2
	OR: Let $AB\parallel DC$ and $AB=DC$, with diagonals meeting at E (1). $\triangle EAB\cong\triangle ECD$ (ASA: alternate angles and $AB=DC$) (1). So $EA=EC$, $EB=ED$: the diagonals bisect each other, so ABCD is a parallelogram (1).	1+1+1
23	$x+y=6$: points $(0,6)$ and $(6,0)$; $x-y=2$: points $(0,-2)$ and $(2,0)$ (1). Drawing the two lines, they cross at $(4,2)$ (1). A unique solution $x=4$, $y=2$, since the lines intersect at one point (1).	1+1+1
	OR: Let the digits be x (tens), y (units). $x+y=11$ (1). $(10x+y)-(10y+x)=27$, so $x-y=3$ (1). Then $x=7$, $y=4$ and the number is 74 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

24	(i) $\frac{3}{4} \times 8 = 6\%$ (1). (ii) Spice $= 0.8$ litre; new volume $= \frac{0.8}{0.06} = 13.33$ litres, so add about 3.33 litres of water (3). (iii) About 13.33 litres (1).	1+3+1
	OR: (i) Seniors: $168 \times 9 = 1512$ cm; juniors: $152 \times 6 = 912$ cm (2). (ii) $\frac{1512+912}{15} = \frac{2424}{15} = 161.6$ cm (2). (iii) The groups are unequal in size, so each average must be weighted by its group size (1).	2+2+1
25	(i) 21 cm $\times$ 7 cm $\times$ 7 cm (1). (ii) $21 \times 7 \times 7 = 1029$ cm^3 (1). (iii) $3 \times \frac{4}{3} \times \frac{22}{7} \times 3.5^3 = 539$ cm^3 (2). (iv) $1029 - 539 = 490$ cm^3 (1).	1+1+2+1
	OR: Cut the curved surface along a line parallel to the axis and open it out: it is a rectangle with sides $2\pi r$ and h, so CSA $= 2\pi rh$ (2). Adding two circular ends, TSA $= 2\pi rh + 2\pi r^2 = 2\pi r(h + r)$ (1). For $r = 7$, $h = 15$: CSA $= 660$ cm^2 (1); TSA $= 660 + 308 = 968$ cm^2 (1).	2+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET B

Full Syllabus (Chapters 9–14) | Class 9 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(a)** the same meaning	interchangeable forms
2	**(a)** Both A and R are true and R is the correct explanation of A	SSS congruence proves the converse
3	**(d)** 3 litres	9×20% = 1.8 litres of solute; total volume = 12 litres
4	**(c)** 16%	$\frac{150\times 30+350\times 10}{150+350}=16\%$
5	**(b)** 75	repeated reduction ends at gcd(75, 0)
6	**(d)** gcd(n, m − n)	every common divisor of m and n also divides m − n
7	**(b)** CD and ∠C	opposite sides share no vertex
8	**(d)** 4.5 cm	PQ = BC/2
9	**(b)** (3, 2)	substitute the pair in both equations
10	**(c)** parallel lines	no common point
11	**(a)** $\pi r(2h+r)$	curved surface plus one base
12	**(a)** 8 times	$V\propto r^3$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	9 is odd, so the 'if' part holds (1). $9=3\times 3$ is not prime, so the 'then' part fails (1).	1+1
14	After the leftmost column a carry of 1 may remain; if not written, a place value is lost (1). E.g. $95+17=112$, but without Step 5 we get 12 (1).	1+1
15	$AG:GD=2:1$ (1); $AG=14$ cm and $GD=7$ cm (1).	1+1
16	Slope $=\frac{y_2-y_1}{x_2-x_1}=\frac{-1-5}{4-(-2)}$ (1) $=\frac{-6}{6}=-1$ (1).	1+1
17	CSA $=2\pi rh=2\times\frac{22}{7}\times 7\times 20$ (1) $=880$ cm^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Justification: $6=3\times 2$, so every multiple of 6 is a multiple of 3 (1). Converse: If a number is a multiple of 3, then it is a multiple of 6 (1). Counterexample: 9 (1).	1+1+1
19	$\angle A=\angle C=x$, $\angle B=\angle D=y$; angle sum $=2(x+y)=360^\circ$ (1) ⇒ $x+y=180^\circ$, i.e. $\angle A+\angle B=180^\circ$ and $\angle B+\angle C=180^\circ$ (1). Co-interior angles add to 180° ⇒ $BC\parallel AD$ and $AB\parallel DC$ (1).	1+1+1
20	Multiply the second by 2: $10x-4y=8$ (1). Add to the first: $13x=26$, so $x=2$ (1). Then $4y=18-6=12$, $y=3$; check $10-6=4$ (1).	1+1+1
21	Water forms a similar cone of height $\frac{1}{3}$ the cup's height (1), so its radius is also $\frac{1}{3}$ (1). Volume ratio $=\left(\frac{1}{3}\right)^3=\frac{1}{27}$ (1).	1+1+1
22	Let the son be s and the father f years: $f=4s$ (1). $f+5=3(s+5)$, so $f-3s=10$ (1). Substituting, $4s-3s=10$, $s=10$ and $f=40$ (1).	1+1+1
	OR: Let apples cost ₹x and grapes ₹y per kg: $3x+2y=540$, $2x+5y=690$ (1). Eliminating x: $(\times 2)\,6x+4y=1080$, $(\times 3)\,6x+15y=2070$, so $11y=990$, $y=90$ (1). Then $3x=540-180=360$, $x=120$. Apples ₹120/kg, grapes ₹90/kg (1).	1+1+1
23	(i) corner cubes: 8 (1). (ii) 12 edges × 2 cubes = 24 (1). (iii) 6 faces × 4 cubes = 24 (½); (iv) inner cube $2\times 2\times 2=8$ (½). Check: $8+24+24+8=64$.	1+1+½+½
	OR: Volume of sphere $=\frac{4}{3}\pi(27)=36\pi$ cm^3 (1). Volume of one cone $=\frac{1}{3}\pi(1)(3)=\pi$ cm^3 (1). Number $=36\pi\div\pi=36$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

24	$AB = 7$, $AC = 24$, $BC = \sqrt{49 + 576} = \sqrt{625} = 25$ (2). $AB^2 + AC^2 = 49 + 576 = 625 = BC^2$, so by the converse the triangle is right-angled at A (1½). Area $= \frac{1}{2} \times 7 \times 24 = 84$ square units (1½).	2+1½+1½
	OR: (i) 49: pairs (1,49), (7,7) → factors 1, 7, 49: 3 (odd) (1). 100: pairs (1,100), (2,50), (4,25), (5,20), (10,10) → 9 factors (odd) (1½). (ii) Converse: If a number has an odd number of factors, then it is a perfect square (1). True: an odd count forces a repeated pair (f, f), so $n = f \times f$ (1½).	2½+2½
25	(i) Combine the scans: check j up to $\max(m, n) = 60$ instead of $m + n = 96$ (2). (ii) Directly compute common divisors, using up to $\min(m, n) = 36$ (2). (iii) Drop lists: keep only most-recent-common-divisor, scanning $k = 2$ to 36 (1).	2+2+1
	OR: Steps: get divisors(m), divisors(n); empty common list; for each x in divisors(m), if x in divisors(n) add to common; report the rightmost element (2). divisors(30) $= [1, 2, 3, 5, 6, 10, 15, 30]$; divisors(45) $= [1, 3, 5, 9, 15, 45]$ (1). Common $= [1, 3, 5, 15]$ (1); gcd $= 15$ (1).	2+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET C

Full Syllabus (Chapters 9–14) | Class 9 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(c)** "If a triangle is equilateral, then all its angles are equal" and "If all angles of a triangle are equal, then it is equilateral"	both directions hold
2	**(b)** not right-angled, since $a^2 + b^2 \neq c^2$	$81 + 144 = 225$ and $16^2 = 256$
3	**(a)** the simple average of the two group averages	the weights are equal
4	**(b)** 72	$\frac{24 \times 68 + 16 \times 78}{24 + 16} = 72$
5	**(b)** 18	repeated reduction ends at gcd(18, 0)
6	**(b)** counting dots needs about 100 times as many dots, but the column method needs only 2 more columns	dots grow with the value, columns grow with the number of digits
7	**(b)** has one internal angle greater than 180°	like the 'DART' shape
8	**(b)** Both A and R are true but R is not the correct explanation of A	both true; the Midpoint Theorem gives both facts, but being parallel does not explain the half-length
9	**(c)** infinite	any value of x gives a value of y
10	**(a)** 2	3 + 2k = 7
11	**(d)** 600 cm^3	$lwh = 12 \times 10 \times 5 = 600$
12	**(c)** 528 cm^2	$2\pi rh = 528$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	Converse: If a number is divisible by 9, then it is divisible by 18 (1). False: 27 is divisible by 9 but not by 18 (1).	1+1
14	$\frac{30 \times 70 + 20 \times 60}{50} = \frac{2100 + 1200}{50} =$ **66** marks	1+1
15	A quadrilateral with one pair of opposite sides both equal and parallel is a parallelogram (2).	2
16	(i) $y = -3$ is horizontal; rise is 0, so slope $= 0$ (1). (ii) $x = 2$ is vertical; run is 0, so the slope is undefined (1).	1+1
17	Volume of cuboid $= 480$ cm^3; volume of one cube $= 8$ cm^3 (1). Number $= 60$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	For $n = 11$: $n^2 + n + 11 = 121 + 11 + 11 = 143$ (1). $143 = 11 \times 13$ (1), so it is composite and the claim is false (1).	1+1+1
19	Guess: slightly above 40 (1). $\frac{20 \times 40 + 61}{21} = \frac{861}{21} = 41$ km (2).	1+2
20	$x + 8y = 140$ and $x + 12y = 200$ (1). Subtracting: $4y = 60$, $y = 15$ and $x = 140 - 120 = 20$ (1). Fare for 20 km $= 20 + 20 \times 15 =$ ₹320 (1).	1+1+1
21	$\pi rl = 462 \Rightarrow r = \frac{462}{\frac{22}{7} \times 21} = 7$ cm (1). Base area $= \frac{22}{7} \times 49 = 154$ cm^2 (1). TSA $= 462 + 154 = 616$ cm^2 (1).	1+1+1
22	$V = \pi r^2 h = \frac{22}{7} \times 0.49 \times 1.5$ (1) $= 2.31$ m^3 (1) $= 2310$ litres (1).	1+1+1
	OR: (i) $V_1 : V_2 = r_1^2 h_1 : r_2^2 h_2 = (1 \times 4) : (4 \times 3)$ (1) $= 1 : 3$ (1). (ii) $\text{CSA}_1 : \text{CSA}_2 = r_1 h_1 : r_2 h_2 = 4 : 6 = 2 : 3$ (1).	1+1+1
23	Converse: If our school has an assembly, then it is Monday (1). The school may also hold assemblies on other days, e.g. on Independence Day (1). So an assembly day need not be Monday; the converse can be false (1).	1+1+1
	OR: Largest side 37: $12^2 + 35^2 = 144 + 1225 = 1369$ (1). $37^2 = 1369$ (1). By the converse of the Baudhāyana–Pythagoras theorem the triangle is right-angled (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) Sum of first collection $= an$, of second $= bm$ (1½). Combined average $= \frac{\text{total sum}}{\text{total count}} = \frac{an+bm}{n+m}$ (1½). (ii) For $n = m$: $\frac{an+bn}{2n} = \frac{n(a+b)}{2n} = \frac{a+b}{2}$ (2).	3+2

	OR: (i) $\frac{46x+36(50-x)}{50}=40$ (1). (ii) $10x=200\Rightarrow x=20$ males; 30 females (2). (iii) $\frac{30\times36+42}{31}=\frac{1122}{31}\approx36.19$ kg (2).	1+2+2
25	Let medians BQ, CP meet at M; X, Y the midpoints of BM, CM (1). $PQ \parallel BC$, $PQ=\frac{1}{2}BC$ and $XY \parallel BC$, $XY=\frac{1}{2}BC$ (Midpoint Theorem in $\triangle ABC$ and $\triangle MBC$) (1). So $\triangle MPQ \cong \triangle MYX$ (ASA) ⇒ $MQ=MX=XB$ and $MP=MY=YC$ (1). Hence $CM:MP=2:1=BM:MQ$ (1). The same argument for BQ and AR gives the same point and ratio ⇒ concurrent (1).	1+1+1+1+1
	OR: PQRS is a parallelogram (Varignon) (1); PR and QS are its diagonals, so they bisect each other (2). If AC = BD then $PQ=\frac{1}{2}AC=\frac{1}{2}BD=QR$, so PQRS is a rhombus (1); the diagonals of a rhombus are perpendicular ⇒ $PR \perp QS$ (1).	1+2+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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